

THE GURVICH–NAUMOVA GM-STRATEGY CONJECTURE FAILS WITH THREE PILES

ABSTRACT. Gurvich and Naumova conjectured that, in the ℓ -player version of exact slow NIM($n, n-1$), the profile in which every player follows the GM strategy is a uniform Nash equilibrium. We disprove the conjecture for every $\ell \geq 3$, already with three piles. From the initial position $(2, \ell+1, \ell+1)$, all-GM play lasts $\ell+2$ moves, whereas player 1 can keep the pile of size 2 on the first move and follow GM thereafter, producing a play of length $\ell+1$. Player 1 remains a winner and improves his payoff by $1/(\ell-1)$. We also show that three is the least possible number of piles for such a counterexample.

1. COUNTEREXAMPLE AND MINIMALITY

Gurvich and Naumova conjectured that the ℓ GM strategies form a uniform Nash equilibrium in the ℓ -player game NIM($n, n-1$) [1, Section 6.5, Conjecture 1], whose statement reads: “The set of ℓ GM-strategies form a (uniform) Nash equilibrium.” (Here and below the numbering is that of the e-print arXiv:2312.08382.) They note that the conjecture holds for $\ell = 2$. We disprove it for every $\ell \geq 3$, already for $n = 3$.

A position of NIM($3, 2$) is a nonnegative triple, written in nondecreasing order. A move keeps one coordinate and decreases the other two by one; the decreased coordinates must therefore be positive. In NIM($n, n-1$) a position is terminal when at least two of its coordinates are non-positive [1, Section 6.5]; for $n = 3$ this says that at most one coordinate is positive, and for $n = 2$ that the only terminal position is $(0, 0)$. Players $1, \dots, \ell$ move cyclically, with player 1 first, and the player whose turn follows the last move is the loser. For a fixed constant C larger than the length of every play from the given initial position, the loser has payoff $L - C$, while each winner has payoff

$$\frac{C - L}{\ell - 1},$$

where L is the length of the play.

For NIM($3, 2$), the GM strategy keeps a smallest coordinate divisible by ℓ , ties being broken in favour of the rightmost such coordinate in the non-decreasing ordering; if no coordinate is divisible by ℓ , it keeps a largest coordinate.

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Theorem 1. *For every integer $\ell \geq 3$, in the ℓ -player game $\text{NIM}(3, 2)$ initialized at*

$$x^0 = (2, \ell + 1, \ell + 1),$$

the all-GM profile is not a Nash equilibrium. Player 1 has a unilateral deviation that increases his payoff by $1/(\ell - 1)$.

Proof. At x^0 , no coordinate is divisible by ℓ , so GM keeps a largest coordinate. At the next position, the coordinate ℓ is the unique multiple of ℓ and is kept. Thereafter the zero coordinate must be kept. Hence all-GM play is

$$\begin{aligned} (2, \ell + 1, \ell + 1) &\longrightarrow (1, \ell, \ell + 1) \longrightarrow (0, \ell, \ell) \\ &\longrightarrow (0, \ell - 1, \ell - 1) \longrightarrow \cdots \longrightarrow (0, 0, 0). \end{aligned}$$

The tail from $(0, \ell, \ell)$ has ℓ moves, so the total length is $L_{\text{GM}} = \ell + 2$. Thus player 3 is the loser and player 1 is a winner.

Let σ^{GM} denote the GM strategy. Define a strategy τ_1 for player 1 that agrees with σ^{GM} at every position except x^0 , where it keeps the pile of size 2. Since $\ell \geq 3$, the coordinate 2 is not a multiple of ℓ , so at $(2, \ell, \ell)$ GM keeps a coordinate ℓ ; at $(1, \ell - 1, \ell)$ the coordinate ℓ is the unique multiple of ℓ and is kept; thereafter the zero coordinate must be kept. Hence, against GM play by all other players, the resulting path is

$$\begin{aligned} (2, \ell + 1, \ell + 1) &\longrightarrow (2, \ell, \ell) \longrightarrow (1, \ell - 1, \ell) \\ &\longrightarrow (0, \ell - 2, \ell) \longrightarrow \cdots \longrightarrow (0, 0, 2). \end{aligned}$$

The tail from $(0, \ell - 2, \ell)$ has $\ell - 2$ moves, so $L_{\text{dev}} = \ell + 1$. Now player 2 is the loser, and player 1 is again a winner. His payoff therefore changes by

$$\frac{C - (\ell + 1)}{\ell - 1} - \frac{C - (\ell + 2)}{\ell - 1} = \frac{1}{\ell - 1} > 0.$$

Both ties encountered above, at x^0 and at $(2, \ell, \ell)$, are between coordinates of equal value, so neither of the two plays depends on the tie-breaking rule. Thus τ_1 is a profitable unilateral deviation. $\square$

The strategies in question do not depend on the initial position: “By definition, the GM-strategies are uniform, that is, independent of x^0 ” [1, Section 6.5]. A Nash equilibrium in GM-strategies would therefore have to hold at every initial position, so the failure at the single position $x^0 = (2, \ell + 1, \ell + 1)$ refutes Conjecture 1 for every $\ell \geq 3$.

The hypothesis $\ell \geq 3$ is used only to ensure that 2 is not a multiple of ℓ (and that player 3 exists). For $\ell = 2$ the GM move at x^0 already keeps the pile of size 2, so τ_1 coincides with the GM strategy and the construction degenerates.

Corollary 2. *Among the games $\text{NIM}(n, n - 1)$ with $n \geq 2$, three is the least number of piles for which the all-GM profile can fail to be a Nash equilibrium.*

Proof. For $n = 2$, the game is $\text{NIM}(2, 1)$. Every move decreases the total number of stones by exactly one, and by the terminality convention above

the only terminal position is $(0, 0)$. Thus every play from (a, b) has length $a + b$, independently of all strategic choices. The loser and all payoffs are consequently fixed, so no unilateral deviation is profitable. Theorem 1 shows that $n = 3$ is attainable. $\square$

REFERENCES

- [1] V. Gurvich and M. Naumova, *Screw discrete dynamical systems and their applications to exact slow NIM*, Discrete Appl. Math. **358** (2024), 382–394; e-print arXiv:2312.08382. doi:10.1016/j.dam.2024.07.018.